MULTI-CLIENT DYNAMIC SDN CHAT NETWORK (PROJECT 8)

Jackfruit Mini Project: TCP Sockets + POX SDN Controller + Mininet **Target: Single Ubuntu VM (20.04 / 22.04 / 24.04)**

1 One-Time Package Installation

Open your Ubuntu terminal and execute system packages & POX clone:

```
sudo apt update && sudo apt install -y mininet
openvswitch-switch xterm python3-pip git
git clone https://github.com/noxrepo/pox.git
~/pox
pip3 install requests --break-system-packages
```

> **Tip:** Allow Mininet to open GUI terminal windows by running: `xhost +SI:localuser:root && xhost +`

2 File Locations & Directory Setup

Organize the 5 project scripts inside your home directory:

- Create folder: `mkdir -p ~/sdn_chat && cd ~/sdn_chat`
- Place **chat_server.py**, **chat_client.py**, **chat_topo.py**, and **benchmark.py** inside `~/sdn_chat/`
- **Crucial:** Copy the controller file into POX's module path:

```
cp sdn_controller.py ~/pox/ext/
```

- Grant execution rights: `chmod +x *.py`

3 Step-by-Step Terminal Execution Workflow

Terminal 1: Start POX SDN Controller

```
cd ~/pox
python3 pox.py forwarding.l2_learning sdn_contr
oller
```

Logs *"REST Policy Bridge listening on http://127.0.0.1:8080"*.

Terminal 2: Launch Mininet Topology

```
sudo mn -c
cd ~/sdn_chat
sudo python3 chat_topo.py
```

Verifies 0% loss on `pingall`, then opens `mininet>` prompt.

Terminal 2 (Inside Mininet CLI): Open Host Windows

```
mininet> xterm h_server h1 h2 h3
```

Four popup windows appear for server & clients. Run in each:

- **In h_server window:** `python3 chat_server.py`
- **In h1 window:** `python3 chat_client.py 10.0.0.100 Alice`
- **In h2 window:** `python3 chat_client.py 10.0.0.100 Bob`
- **In h3 window:** `python3 chat_client.py 10.0.0.100 Charlie`

4 Verifying Rubric Test Scenarios

- **Public Chat & Discovery:** Type `Hello!` on h1; visible to h2 & h3. Test `/rooms` and `/users`.
- **Private Messaging:** On h1, run `/msg Bob secret text`. Only Bob receives it.
- **Dynamic SDN QoS:** On h2, run `/join #emergency`. Observe Terminal 1 install priority flow rule `priority=30000`.
- **Spam Flooding Drop (Firewall):** On h3, send rapid messages. Server invokes REST API; POX pushes hard drop rule `actions=[]` for `10.0.0.3`.
- **Benchmark:** On h1, run `python3 benchmark.py 10.0.0.100` to measure multi-client latency and dump active OVS flows.

5 Quick Troubleshooting & Viva Tips

- **xterm windows don't pop up?** Connect via native terminal tabs using Mininet's namespace bridge:

```
sudo mnexec -a $(pgrep -f "mininet:h_serve
r") bash
```

- **Pingall fails or loops?** Ensure you run `sudo mn -c` before starting Mininet to flush old Open vSwitch kernel datapaths.
- **Controller Module Not Found:** Confirm `ls ~/pox/ext/sdn_controller.py` exists so POX can import it on launch.
- **Viva Key Fact:** TCP byte-stream framing is handled with 4-byte length prefixes (`struct.pack('!I')`), preventing packet fragmentation.